CS50 Web — Lectures 1 & 2 Quiz

Multiple Choice Quiz • 10 Questions

Section A: Git & GitHub (Lecture 1)

1. What is a Git repository?

A) A website used to host HTML pages

B) A file location where all files related to a project are stored

C) A command-line tool for writing code

D) A type of programming language used for version control

2. Which command is used to save a snapshot of the current state of your code in Git?

A) git push

B) git pull

C) git commit -m "message"

D) git clone <url>

3. What happens during a merge conflict in Git?

A) Git automatically deletes the conflicting file

B) Git randomly picks one version of the conflicting changes

C) Two people attempt to change a file in ways that conflict, and Git marks the file to show the conflict

D) The repository is permanently locked until an admin resolves it

4. Which Git command creates a new branch and switches to it at the same time?

A) git branch <name>

B) git checkout -b <name>

C) git merge <name>

D) git reset --hard <name>

5. On GitHub, what is a pull request used for?

A) To download a repository from GitHub to your local machine

B) To delete a forked repository from your account

C) To request that changes from your forked repository be merged into the original repository

D) To revert all commits back to the initial version of the code

Section B: Python (Lecture 2)

6. In Python, which data structure stores key-value pairs and uses curly brackets?

A) List

B) Tuple

C) Set

D) Dictionary

7. What is the purpose of a decorator in Python?

A) To convert a function into a class method

B) To add visual formatting to printed output

C) To modify or extend the behavior of another function using a higher-order function

D) To restrict a function so it can only be called once

8. What does the following Python code do to handle errors?

```
try:
    result = x / y
except ZeroDivisionError:
    print("Error: Cannot divide by 0.")
```

A) It prevents the variable y from ever being zero

B) It catches a division-by-zero error and prints a friendly message instead of crashing

C) It automatically changes y to 1 if it is 0

D) It skips the division entirely regardless of the value of y

9. Which of the following Python sequence types is ordered but immutable?

A) List

B) Set

C) Dictionary

D) Tuple

10. What is the correct way to define a lambda function that doubles a number in Python?

A) lambda x: x * 2

B) def lambda(x): return x * 2

C) lambda: x * 2

D) x => x * 2

Answer Key

Q1: B — A Git repository is a file location where all project-related files are stored. It can be remote (online) or local (on your computer).

Q2: C — The "git commit -m" command takes a snapshot of the current state of your code along with a descriptive message.

Q3: C — A merge conflict occurs when two people change a file in conflicting ways. Git reformats the file to clearly outline the conflict for manual resolution.

Q4: B — "git checkout -b <name>" creates a new branch and switches to it immediately.

Q5: C — A pull request lets you propose changes from your fork to be reviewed and potentially merged into the main repository.

Q6: D — A Python dictionary (dict) stores key-value pairs using curly brackets and colons, e.g. {"Harry": "Gryffindor"}.

Q7: C — A decorator is a higher-order function that wraps another function to modify or extend its behavior, applied using the @ symbol.

Q8: B — The try/except block attempts to run the code inside try, and if a ZeroDivisionError occurs, it catches the exception and runs the except block instead of crashing.

Q9: D — Tuples are ordered (elements have a defined position) but immutable (their elements cannot be changed after creation).

Q10: A — A lambda function uses the syntax: lambda <input>: <output>. So "lambda x: x * 2" takes x as input and returns x * 2.

Lecture 2 Quiz (Python + Basic HTML/CSS)

10 MCQs - compact layout

Answer Key

1) B 2) C 3) B 4) C 5) B 6) B 7) C 8) B 9) B 10) C

Q1. Why can this fail? `num = input('Number: ');  if num > 0:`
[Python - Input & Types]
A) input() returns None
B) input() returns str, compared to int
C) if cannot use variables
D) print() is missing

Q2. Which statement is correct?
[Python - Sequences]
A) Lists are immutable
B) Tuples are mutable
C) Strings are ordered & immutable
D) Sets are ordered & keep duplicates

Q3. After adding 1,2,2,3 to a set, what is `len(s)`?
[Python - Sets]
A) 4
B) 3
C) 2
D) Error

Q4. Which correctly uses `square` from `functions.py`?
[Python - Modules]
A) `print(square(5))`
B) `include functions`
C) `from functions import square`
D) `import square from functions`

Q5. Why does `if not self.open_seats():` work when flight is full?
[Python - OOP]
A) open_seats returns None
B) 0 is False in Python
C) not makes all numbers True
D) passenger count becomes negative

Q6. Which tag is used for the largest heading (default HTML)?
[HTML - Structure]
A) <head>
B) <h1>
C) <header>
D) <title>

Q7. Which attribute sets the destination URL of a link?
[HTML - Links]
A) src
B) link
C) href
D) action

Q8. Which selector targets an element with id='main'?
[CSS - Selectors]
A) .main
B) #main
C) main.
D) *main

Q9. Which property controls space OUTSIDE an element border?
[CSS - Box Model]
A) padding
B) margin
C) border-spacing
D) outline

Q10. Which is the correct way to link an external CSS file?
[HTML/CSS - Inclusion]
A) <style src='styles.css'>
B) <css href='styles.css'>
C) <link rel='stylesheet' href='styles.css'>
D) <script href='styles.css'>

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Section 1: True or False (10 questions)

1. HTML is a markup language that defines the structure of a web page.
2. In CSS specificity, a class selector has higher priority than an id selector.
3. The DOM represents HTML elements using a tree-like structure.
4. Sass allows you to define variables using the '$' symbol.
5. The command 'git clone' is used to upload a local repository to a remote server.
6. In Python, a tuple is an ordered but immutable sequence.
7. The Python 'input()' function automatically converts user input to an integer.
8. In Django, a view is a Python function that takes a request and returns a response.
9. Django's CSRF token is used to prevent Cross-Site Request Forgery attacks.
10. By default, a global variable in views.py is a good way to store per-user data in Django.

Section 2: Multiple Choice Questions (20 questions)

1. **Which HTML tag is used to define the largest heading?**

a) <heading> b) <h6> c) <h1> d) <head>

2. **Which CSS property controls the space inside an element, between its content and its border?**

a) margin b) padding c) border d) spacing

3. **In CSS, how do you select an element with the id 'first-header'?**

a) .first-header b) #first-header c) *first-header d) first-header

4. **What is the correct CSS specificity order from highest to lowest?**

a) element > class > id > inline b) inline > id > class > element c) id > inline > class > element d) class > id > element > inline

5. **Which HTML attribute is used on an tag to provide a description if the image cannot be loaded?**

a) title b) desc c) alt d) label

6. **Which meta tag helps make a page responsive on mobile devices?**

a) <meta name="mobile">

b) <meta name="viewport" content="width=device-width, initial-scale=1.0">

c) <meta name="responsive">

d) <meta charset="UTF-8">

7. **In Bootstrap's grid system, a full row is divided into how many columns?**

a) 6 b) 10 c) 12 d) 16

8. **Which Git command downloads a remote repository to your local computer?**

a) git pull b) git fetch c) git clone d) git download

9. **Which command creates a new branch AND switches to it in one step?**

a) git branch b) git checkout c) git checkout -b d) git switch

10. **Which Git command shows a history of all commits in a repository?**

a) git history b) git log c) git show d) git status

11. **Which Python sequence type is ordered AND mutable?**

a) String b) Tuple c) List d) Frozen set

12. **What does the Python keyword 'self' refer to inside a class method?**

a) The class itself b) The parent class c) The current instance of the object d) A global variable

13. **What does the Python 'input()' function return by default?**

a) An integer b) A float c) A string d) A boolean

14. **Which Python keyword is used to define a function?**

a) function b) def c) func d) lambda

15. **Which Python exception is raised when you try to divide a number by zero?**

a) ValueError b) TypeError c) ZeroDivisionError d) ArithmeticError

16. **In Django, which file is used to map URL patterns to views?**

a) views.py b) models.py c) urls.py d) settings.py

17. **Which Django function is typically used to render a template with context?**

a) HttpResponse() b) render() c) redirect() d) template()

18. **Which method is typically used for form submissions that may change the server state?**

a) GET b) POST c) PUT d) HEAD

19. **What is the purpose of '{% csrf_token %}' in a Django form?**

a) To style the form b) To protect against Cross-Site Request Forgery c) To encrypt passwords d) To validate email fields

20. **Which Django feature allows storing unique data per visitor on the server side?**

a) Cookies b) Sessions c) Context d) Templates

Section 3: Extended Matching Questions (15 questions)

Options:

A. B. C. <form> D. flexbox E. Bootstrap F. @media G. Sass variable H. git commit I. git push J. git merge K. Branching L. List M. Tuple N. Dictionary O. Set P. lambda Q. try / except R. render() S. Template T. Sessions

1. An HTML element used to create a list where the order of items does not matter.
2. An HTML element used to collect input from users.
3. A CSS feature used to change styling based on the screen size of the device.
4. A front-end library that provides a 12-column grid system and pre-styled components.
5. A Sass feature declared with '$' that lets you reuse a value such as a color.
6. The Git command that takes a local snapshot of the current state of your code.
7. The Git command used to publish local commits to a remote GitHub repository.
8. A Git workflow that lets you develop a new feature separately from the main code base.
9. An ordered and mutable Python sequence used to store any variable types.
10. A Python data structure that stores unique, unordered elements (no duplicates).
11. A Python key-value data structure that is unordered and mutable.
12. A one-line anonymous Python function, useful as a 'key' argument to 'sort'.
13. A Python construct used to catch and handle runtime errors such as ZeroDivisionError.

14. A Django function that loads an HTML template and returns it with optional context.
15. A Django feature that lets multiple HTML files share a common layout via '{% block %}'.

Section 4: Short Answer Questions (Answer 2 only)

1. Briefly explain the difference between an HTML 'id' and a 'class', and how each is selected in CSS.
2. What is a merge conflict in Git, and in what situations does it typically occur?
3. Explain the difference between mutable and immutable sequences in Python, and give one example of each.
4. What is the difference between a static and a dynamic website, and how does Django help us build dynamic ones?
5. What are Django sessions, and why are they useful compared to storing data in a global variable in views.py?